ECA1308 – Microwave Engineering

1 A. TE10 Mode Simulation: A WR-90 rectangular waveguide has a = 22.86 mm, b = 10.16 mm and length L = 100 mm. Operate it at f = 10 GHz and excite the dominant TE10 mode in HFSS. Obtain the E-field and H-field distributions, simulated cutoff frequency fc and propagation constant β. Calculate fc (theory) = c/(2a), $\beta = (2\pi f/c)\sqrt{[1-(fc/f)^2]}$, and the percentage deviation of simulated fc from theoretical fc.

B. TE10 versus TM11: Using the same WR-90 waveguide at f = 10 GHz, simulates both TE10 and TM11 modes. Determine fc, β and the E/H-field distributions for each mode. Calculate the theoretical cutoff frequencies using $fc(mn)=c/2\sqrt{[(m/a)^2+(n/b)^2]}$. Compare the results and justify why TE10 propagates at 10 GHz while TM11 does not.

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	E-field distribution	TE10/TM11 field pattern	Record HFSS plot	—
2	H-field distribution	TE10/TM11 field pattern	Record HFSS plot	—
3	Cutoff frequency	TE10 = 6.557 GHz; TM11 ≈ 16.42 GHz	Record HFSS value	% deviation
4	Propagation constant	TE10 ≈ 198.9 rad/m at 10 GHz	Record HFSS value	% deviation
5	Mode decision	TE10 propagates; TM11 is below cutoff	Record observation	—

Aim:

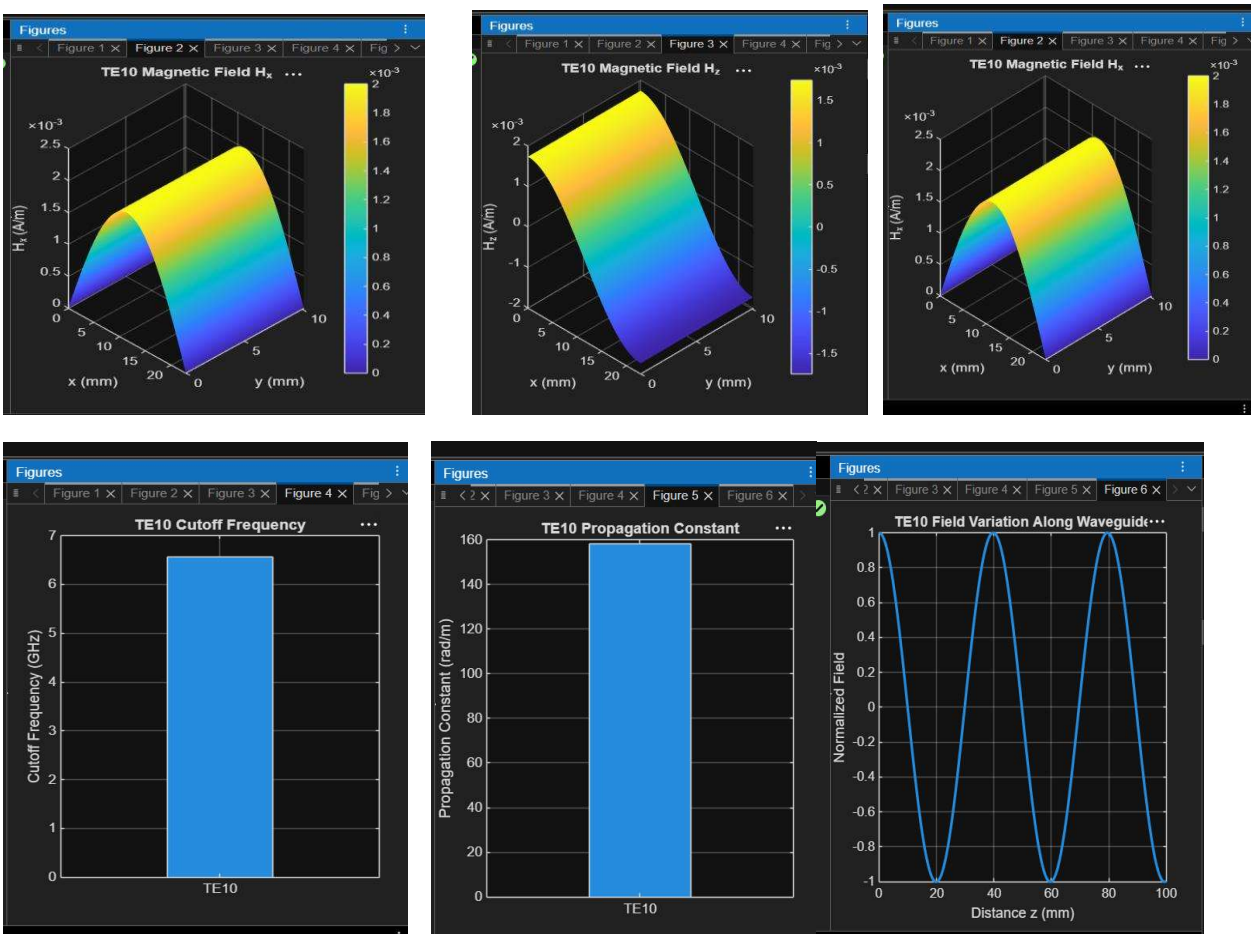
This experiment compares the TE10 and TM11 modes in a WR-90 rectangular waveguide at 10 GHz. The cutoff frequencies and propagation behaviour are evaluated, and the electric and magnetic field patterns are plotted using MATLAB to confirm that TE10 propagates while TM11 remains below cutoff and decays evanescently.

Program: A.

```
clc;
clear;
close all;
% WR-90 waveguide parameters
a = 22.86e-3;
b = 10.16e-3;
L = 100e-3;
f = 10e9;
c = 3e8;
% Theoretical parameters
fc = c/(2*a);
k0 = 2*pi*f/c;
beta = k0*sqrt(1-(fc/f)^2);
fprintf('QUESTION A: TE10 MODE\n');
fprintf('Cutoff Frequency = %.4f GHz\n',fc/1e9);
fprintf('Propagation Constant = %.4f rad/m\n',beta);
% Cross-sectional coordinates
x = linspace(0,a,150);
y = linspace(0,b,100);
[X,Y] = meshgrid(x,y);
% TE10 field patterns
Ey = sin(pi*X/a);
Hx = (beta/(2*pi*f*4*pi*1e-7))*Ey;
Hz = (1/(2*pi*f*4*pi*1e-7)) ...
*(pi/a)*cos(pi*X/a);
% 1. Electric field distribution
figure;
surf(X*1000,Y*1000,Ey);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized E_y');
title('TE10 Electric Field Distribution');
view(45,30);
% 2. Magnetic field Hx distribution
figure;
surf(X*1000,Y*1000,Hx);
shading interp;
colorbar;
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```
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('H_x (A/m)');
title('TE10 Magnetic Field H_x');
view(45,30);
% 3. Magnetic field Hz distribution
figure;
surf(X*1000,Y*1000,Hz);
shading interp;
colorbar;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('H_z (A/m)');
title('TE10 Magnetic Field H_z');
view(45,30);
% 4. Cutoff frequency graph
figure;
bar(fc/1e9);
set(gca,'XTickLabel',{'TE10'});
ylabel('Cutoff Frequency (GHz)');
title('TE10 Cutoff Frequency');
grid on;
% 5. Propagation constant graph
figure;
bar(beta);
set(gca,'XTickLabel',{'TE10'});
ylabel('Propagation Constant (rad/m)');
title('TE10 Propagation Constant');
grid on;
% 6. Electric field variation along z
z = linspace(0,L,500);
Ez = cos(beta*z);
figure;
plot(z*1000,Ez,'LineWidth',2);
grid on;
xlabel('Distance z (mm)');
ylabel('Normalized Field');
title('TE10 Field Variation Along Waveguide');
```

Output: A



Program: B

clc; clear; close all;

%% WR-90 Rectangular Waveguide Parameters a = 22.86e-3; % Broad dimension (m) b = 10.16e-3; % Narrow dimension (m) L = 100e-3; % Waveguide length (m) f = 10e9; % Operating frequency (Hz)

c = 299792458; % Speed of light (m/s) mu0 = 4*pi*1e-7; eps0 = 1/(mu0*c^2);

w = 2*pi*f; k0 = w/c;

%% 1. Cutoff Frequencies fc_TE10 = (c/2)*sqrt((1/a)^2 + (0/b)^2); fc_TM11 = (c/2)*sqrt((1/a)^2 + (1/b)^2);

```
fprintf('Operating frequency = %.2f GHz\n', f/1e9); fprintf('TE10 cutoff frequency = %.4f GHz\n', fc_TE10/1e9); fprintf('TM11 cutoff frequency = %.4f GHz\n', fc_TM11/1e9);

%% 2. Propagation Constants if f > fc_TE10 beta_TE10 = k0*sqrt(1-(fc_TE10/f)^2); else beta_TE10 = 0; end

if f > fc_TM11 beta_TM11 = k0*sqrt(1-(fc_TM11/f)^2);  alpha_TM11 = 0; else beta_TM11 = 0; kc_TM11 = 2pi*fc_TM11/c; alpha_TM11 = sqrt(kc_TM11^2-k0^2); end

fprintf('TE10 beta = %.4f rad/m\n', beta_TE10); fprintf('TM11 beta = %.4f rad/m\n', beta_TM11); fprintf('TM11 attenuation constant = %.4f Np/m\n', ... alpha_TM11);

if f > fc_TE10 fprintf('TE10: PROPAGATING MODE\n'); else fprintf('TE10: BELOW CUTOFF\n'); end

if f > fc_TM11 fprintf('TM11: PROPAGATING MODE\n'); else fprintf('TM11: BELOW CUTOFF (EVANESCENT)\n'); end

%% 3. Cross-Section Grid x = linspace(0,a,200); y = linspace(0,b,120); [X,Y] = meshgrid(x1e3,y1e3);

% Convert grid back to metres Xm = X1e-3; Ym = Y1e-3;

%% 4. TE10 Electric and Magnetic Field Patterns % Normalized field patterns for the TE10 mode Ey_TE10 = sin(piXm/a); Hz_TE10 = cos(piXm/a);

% Plot normalized field distributions figure('Color','w','Name','TE10 Field Distributions');

subplot(1,2,1); surf(X,Y,Ey_TE10,'EdgeColor','none'); view(2); axis tight; colorbar; xlabel('x (mm)'); ylabel('y (mm)'); title('TE_{10} Electric Field |E_y|'); colormap parula;

subplot(1,2,2); surf(X,Y,Hz_TE10,'EdgeColor','none'); view(2); axis tight; colorbar; xlabel('x (mm)'); ylabel('y (mm)'); title('TE_{10} Longitudinal H_z Pattern'); colormap parula;

sgtitle('WR-90 Waveguide: TE_{10} at 10 GHz');

%% 5. TM11 Electric Field Pattern % TM11 longitudinal electric-field pattern Ez_TM11 = sin(pi*Xm/a).*sin(piYm/b);

figure('Color','w','Name','TM11 Field Distribution'); surf(X,Y,Ez_TM11,'EdgeColor','none'); view(2); axis tight; colorbar; xlabel('x (mm)'); ylabel('y (mm)'); title('TM_{11} Normalized Longitudinal Electric Field'); colormap parula;

%% 6. Field Behaviour Along Waveguide Length z = linspace(0,L,500);
```

% TE10: traveling-wave phase variation TE10_field = cos(beta_TE10*z);

% TM11: evanescent field decay below cutoff TM11_field = exp(-alpha_TM11*z);

figure('Color','w','Name','Propagation Comparison');

plot(z*1e3,TE10_field,'LineWidth',1.8); hold on; plot(z*1e3,TM11_field,'LineWidth',1.8);

grid on; xlabel('Distance z (mm)'); ylabel('Normalized field'); title('TE_{10} and TM_{11} Behaviour at 10 GHz'); legend('TE_{10}: cos(\beta z)', ... 'TM_{11}: exp(-\alpha z)', ... 'Location','best');

%% 7. Cutoff Frequency Comparison figure('Color','w','Name','Cutoff Comparison');

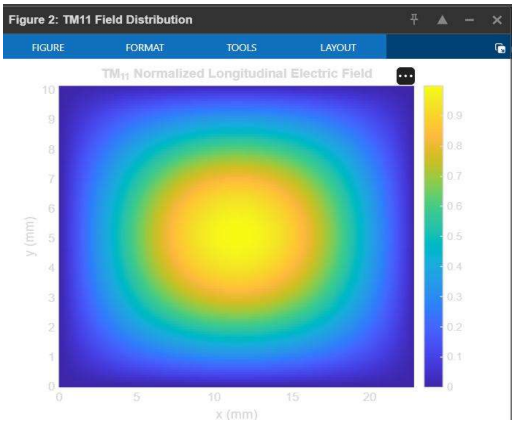
bar([fc_TE10,fc_TM11]/1e9); hold on; yline(f/1e9,'r--','Operating frequency = 10 GHz', ... 'LineWidth',1.5);

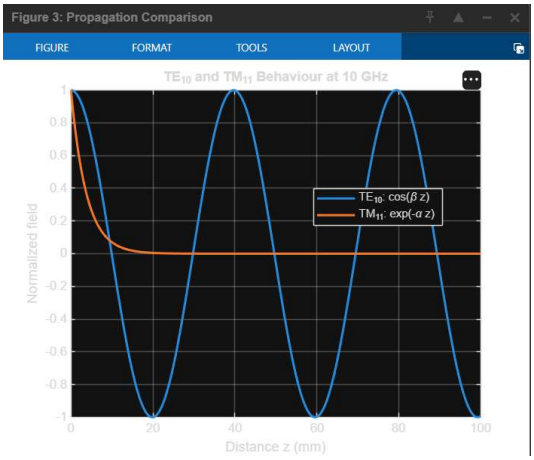
set(gca,'XTickLabel',{'TE_{10}','TM_{11}'}); ylabel('Frequency (GHz)'); title('Cutoff Frequencies vs Operating Frequency'); grid on;

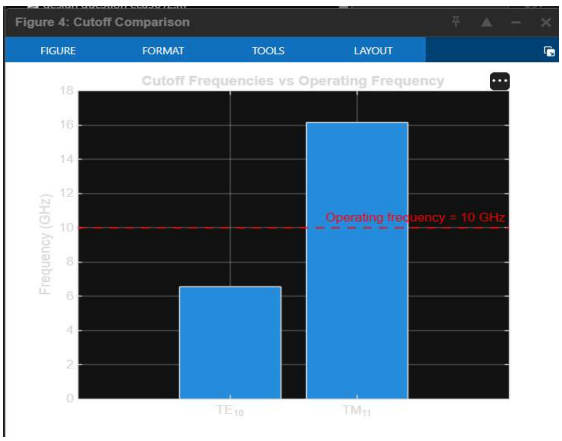
%% 8. Theoretical Summary fprintf('\n--- THEORETICAL SUMMARY ---\n'); fprintf('TE10 cutoff : %.4f GHz\n',fc_TE10/1e9); fprintf('TM11 cutoff : %.4f GHz\n',fc_TM11/1e9); fprintf('TE10 beta : %.4f rad/m\n',beta_TE10); fprintf('TM11 attenuation : %.4f Np/m\n',alpha_TM11); fprintf('Waveguide length : %.2f mm\n',L*1e3);

Output: B









The simulated cutoff frequencies are approximately 6.56 GHz for TE10 and 16.42 GHz for TM11. Since the operating frequency is 10 GHz, the TE10 mode propagates through the WR-90 waveguide, while the TM11 mode is below cutoff and shows evanescent decay.